convenience, but we don’t do that carving randomly. There are right and wrong ways to divide the wave function into branches, and the right ways leave us with independent worlds that obey approximately classical laws of physics. Which ways actually work is ultimately determined by the fundamental laws of nature, not by human whimsy.

Emergence is not a generic feature of physical systems. It happens when there’s a special way of describing the system that involves much less information than a complete description would, but nevertheless gives us a useful handle on what’s going on. That’s why it makes sense for us to carve up reality in the way we do, describing tables and chairs and branches of the wave function.

Think of a planet orbiting the sun. A planet like the Earth contains roughly 10^{50} particles. To describe the state of the Earth exactly, even at the classical level, would require listing the position and momentum of every one of those particles, something that is beyond even our wildest imagination of supercomputing power. Happily, if what we care about is just the orbit of the planet, the vast majority of that information is completely unnecessary. We can instead idealize the Earth as a single point, located at the Earth’s center of mass and with the same total momentum. The state of this idealized point is specified by a position and momentum, and that very tiny amount of information (six numbers, three each for position and momentum, as opposed to 6×10^{50} numbers, positions and momenta for each particle) is all we need to calculate its trajectory. That’s emergence: a way of capturing important features of a system using far less information than an exhaustive description would entail.*

We often talk about emergent descriptions in terms of how “convenient” they are for us to use, but don’t be tricked into thinking there’s anything anthropocentric going on. Tables and chairs and planets would still exist even if there were no human beings to talk about them. “Convenience” is a shorthand for indicating an objective physical property: the existence of an accurate model of the system that requires only a tiny fraction of the full information characterizing it.